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Continuous Assessment Test (CAT) – I – July 2026

Programme	:	B.Tech	Semester	:	Fall Semester 2026-27
Course Title	:	Database Management System	Course Code	:	CSE3001
Date/ Session	:	**10-8-2026 / S-I**	Slot	:	A11+A12+A13
Time	:	1 ½ hours	Max. Marks	:	50

Answer all the Questions

Q.No.	Sub. Sec.	Question Description	Marks
1		Explain the three-schema architecture of a database system. Describe the External, Conceptual, and Internal levels with a neat diagram and discuss their significance	10
2		Construct an E-R diagram for a Hospital Management System consisting of Doctors, Patients, Departments, and Appointments. A patient can book multiple appointments with different doctors, and each doctor belongs to one department. Entities • Doctor (Doctor_ID, Name, Specialization, Salary) • Patient (Patient_ID, Name, Age, Gender, Phone) • Department (Dept_ID, Dept_Name, Location) • Appointment (Appointment_ID, Appointment_Date, Diagnosis) Relationships • Patient — Appointment: Books • Doctor — Appointment: Attends • Department — Doctor: Employs Construct the relational tables with attributes, primary keys, and foreign keys.	10
3		Consider the relations STUDENT (StudentID, Name, DeptID), DEPARTMENT (DeptID, DeptName) and RESULT (StudentID, Subject, Marks). Write Relational Algebra expressions to find: (a) Students belonging to the CSE department. (b) Students scoring more than 80 marks in DBMS. (c) Names of students who have not appeared in DBMS.	10
4		Consider the relation EMP(EmpID, EmpName, DeptNo, DeptName, ProjectID, ProjectName) with the functional dependencies: EmpID → EmpName, DeptNo, ProjectID DeptNo → DeptName ProjectID → ProjectName Identify the highest normal form of the relation and normalize it up to BCNF.	10
5		Explain the concept of Keys in the Relational Model. Differentiate between Super Key, Candidate Key, Primary Key, Alternate Key, Composite Key, and Foreign Key with suitable examples	10

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